DARDI PEKA

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Eligible for the UK Graduate visa after the MSc ends in September 2027

PROFILE

MSc Advanced Aeronautical Engineering student at Imperial College London with a First-Class BEng in Aerospace Engineering at Queen Mary University of London. As Head of Engineering at Queen Mary Formula Student, led 50+ members to the team's first fully assembled car in over two years, having deputised as Chief Engineer at FSUK 2025. Experienced in design, simulation and vehicle dynamics. Seeking a graduate engineering opportunity.

EDUCATION

Imperial College London - London, UK
Sept 2026 – Sept 2027

MSc Advanced Aeronautical Engineering

- Modules include: Applied Aerodynamics, Turbulence and Turbulence Modelling, Advanced Propulsion, Aerospace Structures, Aircraft Operations, Composite Innovation, Analytical Characterisation of Composite Materials

Queen Mary University of London - London, UK
Sept 2023 – Jul 2026

BEng Aerospace Engineering - First Class (76.0%)

- Key modules: Fluid Mechanics and Thermodynamics (97%), Numerical Methods and Data Science (92%), Subsonic Aerodynamics and Wings (83%), Aerodynamics of High-Speed Flows (82%), Conceptual Design of Aircraft (78%)

LEADERSHIP EXPERIENCE

Queen Mary Formula Student - London / Towcester, UK
Sept 2023 – Aug 2026

Head of Engineering
Jul 2025 – Aug 2026

- Led 7 department heads, 7 deputies and 40+ members, working with a leadership team of two, to the team's first fully assembled car in over two years, running weekly integration meetings against an engineering Gantt chart.
- Re-planned the build after a lab closure and procurement delays cost two months, agreeing a revised goal with the leadership team. Delivered the car to full scrutineering at FSUK 2026.
- Resolved packaging conflicts between five subsystems by having department heads justify each decision against the team goal. Maintained the master CAD assembly and reviewed engineering drawings.
- Supervised and helped carry out the wiring loom rebuild, after which the engine ran for the first time in a year; contributed to the rear suspension hardpoint redesign and engine bay repackaging.
- Wrote the Engineering Design Report (scored higher than the previous Formula Class entry), presented the overall vehicle concept to the design judges and led the largest engineering handover to date.

Head of Aerodynamics
Jul 2024 – Jul 2025

- Deputised as Chief Engineer at FSUK 2025 while Head of Aerodynamics, leading the team to 9th in design, 5th in cost and 7th overall of 40+ teams in the concept class.
- Designed a new IC25 front wing with STAR-CCM+ simulation and parametric SolidWorks CAD, predicting 2.2x the downforce at 30% lower component cost and 4.5 kg less mass.
- Managed a 6-person department: trained new members in STAR-CCM+ and Ansys Fluent, then assigned and guided their simulation work, and provided simulation support to other departments such as powertrain.
- Organised the department's bill of materials and competition documentation, and wrote a detailed handover for the department deputy.

Senior Team Member - IC Aerodynamics & Lap Time Simulation
Nov 2023 – Jul 2024

- Co-presented the suspension and chassis design to the judges with the Head of Suspension (concept class, top 5).
- Developed front wing concepts within the department, creating the best design for the concept submission. Worked on the team's electric vehicle lap time model in IPG CarMaker and OptimumLap (ranked top 7).

Associate Team Member
Sept 2023 – Nov 2023

- Completed 4-week CAD, FEA and CFD training, an aero package review and the department task while assisting other associates; joined the Senior Team within two months.

ENGINEERING PROJECTS

IC26 Ride and Damping Study - Individual
Aug 2026 – Sept 2026

Self-initiated study of ride and grip on the QMFS IC26 Formula Class car.

- Derived the suspension equations by hand, then built simple and detailed models in MATLAB and Simulink and verified both against the hand calculations.
- Examined how tyre lift-off over kerbs lowers the grip-optimal damping below the simple model's prediction.
- Retested the study on measured tyre data: headline result confirmed, two secondary conclusions revised and published.

Integrated Tiltrotor Design Project - Queen Mary University of London
Sept 2025 – May 2026

Team of 6 modernising the Bell XV-15 for urban air mobility; individual workstream on conversion and wake aerodynamics.

- Reviewed the literature to scope the individual workstream and wrote the front-end design report.
- Built and verified a transient CFD model of the nacelle conversion in STAR-CCM+, showing drag rising sharply beyond about 30 degrees of tilt.
- Selected the drivetrain, which the wing design accounted for. The wake results guided the empennage work.
- Presented the project's societal, commercial and risk case, commended by the supervisor and external examiner, and defended the report work in an individual viva; project marked 76%.

Steering Wheel Design and Manufacture - Queen Mary Formula Student
Feb 2024 – Apr 2026

Owned from concept to car over three years: designed for IC24, carried over to IC25, rebuilt for IC26.

- Designed the wheel from scratch to clear the front hoop and improve steering torque and ergonomics within the rules, prototyping the grips with the drivers.
- Evaluated carbon fibre against aluminium with a weighted decision matrix, Ansys Workbench FEA and a DFMEA.
- Rebuilt it for IC26 by resin infusion with the Deputy and Head of Composites after an early wet-layup, and helped document the process in a write-up that became teaching material for the department.

Supersonic Wind Tunnel Experiment - Queen Mary University of London
Mar 2026 – Apr 2026

Individual coursework: schlieren analysis of a double-wedge aerofoil at Mach 1.4.

- Measured shock and expansion angles from schlieren images with 95% confidence intervals, automating frame selection with a MATLAB script that picks the clearest frames.
- Determined where theory broke down and formulated a possible solution to the asymmetrical flow; report marked 74%.

OTHER EXPERIENCE

InvestIN Young Engineer Experience / 2Excel Engineering - London / Lasham, UK
Aug 2022

- Networked with Rolls-Royce and BAE Systems engineers during design challenges.
- Collaborated with a group of 12 to assemble a glider and observed its launch. Investigated how an MRO facility runs maintenance, safety and testing across aircraft with varying mission profiles.

SKILLS

- **CFD Simulation:** Simcenter STAR-CCM+, Ansys Fluent, RANS/URANS, overset grids, SST k-ω
- **CAD Design & FEA:** SolidWorks modelling, DFMEA, Ansys Workbench
- **Vehicle Simulation & Dynamics:** MATLAB and Simulink, IPG CarMaker, OptimumLap
- **Post-Processing & Analysis:** MATLAB (PIVlab), FFT / frequency-domain analysis, uncertainty analysis
- **Manufacturing:** Wet layups, resin infusion, 3D printing for prototyping

CERTIFICATES AND MEMBERSHIPS

- CSWA - SolidWorks CAD Design Associate; IBM Enterprise Design Thinking Practitioner
- IMechE Student Member

INTERESTS AND ACTIVITIES

Sketching cars and aircraft; scale models (Gundam, car and aircraft kits); flight simulators; Sunday league basketball.